



JACQUELINE AURIOL

FRENCH AVIATOR Jacqueline Auriol (1917–2000) was one of the few women to trespass into the macho territory of the jet test pilot. When she took up flying in the late 1940s, Auriol was already a minor celebrity, as the glamorous daughter-in-law of the president of the French Republic. She was gaining a reputation as a competition flier when, in 1949, a hydroplane in which she was traveling crashed in the Seine River. Auriol suffered serious facial injuries and underwent 22 operations to restore her features.

Undeterred, Auriol continued flying and qualified to pilot jets through a military training course in the United States. In May 1951, she flew a British Vampire at 511mph (818kph), a new record for a woman. From then until 1964, she and American pilot Jacqueline Cochran vied for the title of “fastest woman in the world.” Both Jacquelines broke the sound barrier in 1953, Cochran first in an F-86 Sabre, then Auriol on board a Dassault Mystère II. By 1959 the informal competition was being conducted at speeds above Mach 2. Auriol set her last record in June 1963, when she flew a Dassault Mirage III R at 1,274mph (2,039kph). Cochran topped this in 1964, emerging as the overall winner.



SUPERSONIC JACQUELINE

Jacqueline Auriol, France's leading woman pilot of the postwar era, relaxes in her flying suit in front of a Dassault Mirage III R, the aircraft in which she reached a record-breaking speed of 1,274mph (2,039kph) in 1963.

Supersonic jets

Jet fighters were soon catching up with the rocket aircraft for speed and altitude. Piloted by George Welch, the prototype of the F-86 Sabre passed Mach 1 in a dive as early as April 1948. The F-100 Super Sabre, first flight-tested at Edwards Air Force Base in 1953, was the first jet fighter designed to go supersonic in level flight. By 1958, the F-104 Starfighter was able to set a jet speed record of 1,404mph (2,259kph) and fly to an altitude of over 100,000ft (30,500m). In fact, by the end of the 1950s, Mach 2 in level flight could be regarded as a standard requirement for state-of-the-art interceptors – not only the American airplanes, but aircraft such as the Soviet MiG-21, the French Dassault Mirage III, and the British Lightning.

Turbojet engines were given the potential for short bursts of very high power through the use of afterburners – spraying fuel into the hot exhaust gases to give an extra kick. And dazzlingly innovative airframes were designed to minimize drag and maximize performance. The Mirage adopted a delta-wing that gave a large wing surface for a small aircraft; the MiG-21 was a “tailed” delta-wing aircraft. At the other extreme, the F-104 – habitually described as “the missile